

Corolario de separación de puntos en un espacio normado

Corolario 1. Sea X un espacio normado y sean $x_1, x_2 \in X : x_1 \neq x_2$. Entonces,

$$\exists L \in X^* : \|L\| = 1 \wedge L(x_1) \neq L(x_2).$$

Demostración: Tomamos $x_0 = x_1 - x_2 \neq 0$ y aplicamos el teo-esp-normado-separacion-punto-cero/ para obtener $L \in X^*$ tal que $\|L\| = 1$ y $L(x_0) = \|x_0\| \neq 0$. Por tanto,

$$L(x_1) - L(x_2) = L(x_1 - x_2) = L(x_0) \neq 0 \implies L(x_1) \neq L(x_2).$$

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